Restore-Sensitive Support Normal Forms for Multi-Tag Quantum Compilation

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Abstract

Support bounds in exact quantum compilation often arise from a restricted deletion proof rather than from an intrinsic lower bound of the compiler. We isolate that distinction for a MultiTag-TARE grammar. The combinatorial zero-sum invariant used in the proof is donor mathematics: for a finite abelian signature group `H` and a fixed allowed alphabet `A`, let $\mathrm{zsf}(H; A)$ be the longest zero-sum-free sequence over `A`. The new content is the compiler-side contract that makes this donor invariant usable.

We state the deletion theorem with the quantifiers needed by a real compiler: each generator has an instance-level alphabet fixed before optimization; a legal deletion preserves feasibility of the whole instance; and repeated deletions cannot increase any generator's support. Starting from an optimum and applying admissible zero-sum deletions until a global fixed point is reached gives one optimum satisfying the zsf ceiling simultaneously for every constrained generator. In an elementary binary quotient this implies the familiar rank ceiling, but that rank statement is explicitly binary-specific and is not itself novel.

For the registered arbitrary-block MultiTag grammar, one frame-support deletion changes one argument of one charged local `b`-way Restore term. The exact one-argument sensitivity of that term is `b-1`; therefore the objective-deletion cone is

$mu \geq (b-1)t_R$.

Inside this cone, every admitted instance has an optimum with

$$\mathrm{support}(R) \leq \mathrm{zsf}(H_R; A_R) \leq \mathrm{rank}(H_R) \leq s+1$$

for every frame `R`, where `A_R` is the fixed instance-level signature alphabet and $H_R =< A_R >$. The frozen one-Tag, three-block R6M family is the sharp control: the certificate ceiling is two and independent compiler evidence establishes intrinsic support $kappa_R 6M = 2$. We do not claim general MultiTag sharpness, necessity outside the cone, or physical quantum advantage.

Keywords: quantum compilation; Pauli strings; support normal forms; zero-sum deletion; sparse optimization

1 1. Scientific question and novelty boundary

The question is not whether Davenport-type zero-sum thresholds exist; they do, and that mathematics is prior work. The question is what a compiler must establish before such a threshold becomes a valid simultaneous normal form under its semantics and objective.

The surviving paper-level contributions are therefore:

1. an explicit whole-instance deletion contract and global fixed-point proof that yields simultaneous support ceilings;
2. the exact Restore sensitivity `b-1` and resulting objective cone for the stated MultiTag grammar;

3. the compiler-specific separation between certificate ceiling and intrinsic support, with R6M as a sharp control.

The following are **not** claimed as novel: Davenport constants, zero-sum-free sequence theory, the binary rank corollary, generic proof-system-relative complexity, or sparse-support theory.

2 2. Fixed alphabets and zero-sum-free sequences

Let `H` be a finite abelian group written additively and let `A subseteq H` be a finite alphabet fixed independently of the optimum being bounded. A **subsequence** means an arbitrary sub-multiset of positions; it need not be contiguous.

A sequence over `A` is zero-sum-free if no nonempty subsequence has sum zero. Define $\mathrm{zsf}(H; A) = max(0 \cup |W| : Wiszero-sum-freeoverA)$.

The explicit zero in the maximum handles degenerate alphabets. For the full nonzero alphabet the object reduces to the standard small-Davenport threshold; for restricted alphabets it is an alphabet-specific variant. We use zsf only as notation for this donor-owned combinatorial object.

3 3. Simultaneous deletion theorem

Fix one optimization instance. For each constrained generator `R`, let `A_R` be the set of signatures realizable by any admissible local state of that instance, fixed before optimization, and let every active coordinate of `R` carry `v_q in A_R subseteq H_R`.

Assume:

1. **nonzero total:** every feasible constrained generator satisfies $sum_q v_q \neq 0$;
2. **whole-instance deletion soundness:** deleting any nonempty zero-sum subsequence of coordinates of `R` preserves all constraints of the full instance, not only constraints local to `R`;
3. **objective dominance:** the same deletion does not increase the objective;
4. **support monotonicity:** deleting coordinates of one generator does not activate coordinates of another generator or otherwise increase any generator's support.

Theorem 1 (simultaneous alphabet ceiling). Every admitted instance has an exact optimum such that, simultaneously for every constrained generator `R`,

$\mathrm{support}(R) \leq \mathrm{zsf}(H_R; A_R)$.

Proof. Start from any optimum. If some generator `R` has a signature sequence longer than $\mathrm{zsf}(H_R; A_R)$, it contains a nonempty zero-sum subsequence. Since the total signature of `R` is nonzero, that subsequence cannot be the whole active sequence. Delete it. Assumptions 2 and 3 preserve feasibility and optimality, while total support strictly decreases and no other support increases by Assumption 4. Repeat whenever any generator remains reducible. Total support is a nonnegative integer, so the process terminates at a global fixed point. At that fixed point no generator contains a nonempty zero-sum subsequence; hence every generator's active signature sequence is zero-sum-free over its fixed alphabet and has length at most $\mathrm{zsf}(H_R; A_R)$. This one terminal optimum satisfies all generator bounds simultaneously. □

This global fixed-point argument is the required quantifier step; a one-pass instruction to "repeat for every generator" is not sufficient when generators may share constraints.

4 4. Binary rank is only a corollary

If $H = F_2^d$, any sequence of more than `d` vectors is linearly dependent. The dependent positions form a nonempty zero-XOR subsequence, so

zsf$(F_2^d; A) \leq d$.

If `A` contains a basis, equality holds for the abstract deletion language.

This statement is specific to elementary binary groups. It must not be generalized to arbitrary finite abelian groups: for example, in `Z_n` with alphabet `{1}`, the sequence of `n-1` ones is zero-sum-free even though the group has rank one.

A rank ceiling becomes an intrinsic compiler statement only after a separate compiler lower witness.

5 5. MultiTag-TARE grammar and the nonzero-total invariant

Fix $b \geq 2$ ordered blocks and $s \geq 0$ shared Tag Paulis `S_1,...,S_s`. Each constrained frame Pauli `R` has an anticommuting partner `R'`. At an active frame coordinate `q`, define

$v_q = (< R_q, R'_q >, < S_1 q, R_q >, ..., < S_s q, R_q >) in F_2^(s + 1)$.

The XOR of the first components equals the global binary symplectic product `<R,R'>`. Since `R` and `R'` anticommute, $< R, R' \geq 1$, and therefore the total signature is nonzero. This discharges Theorem 1's first assumption for the registered grammar.

For each frame `R`, define `A_R` at the instance level as the set of signatures realizable by any admissible local frame state, not the alphabet observed in a particular optimum. Set $H_R =< A_R >$.

6 6. Exact Restore sensitivity

The local Restore term is

$$F_b(a_1, ..., a_b) = 1$$

when all `b` letters are the same nonidentity Pauli, and otherwise equals the number of nonidentity letters.

Lemma 2 (one-argument sensitivity). Replacing one argument of `F_b` can increase its value by at most `b-1`, and the bound is attained whenever the local Pauli alphabet contains at least two distinct nonidentity letters.

Proof. Away from the all-equal nonidentity state, changing one argument changes ordinary nonidentity Hamming weight by at most one. Entering the special state lowers the value. Leaving the all-equal nonidentity state by replacing one letter with a different nonidentity letter changes the value from `1` to `b`, giving the exact increase `b-1`. □

The registered MultiTag incidence contract assigns a frame-support deletion to exactly one argument replacement in exactly one charged local `F_b` term. This incidence is part of the grammar; without it the following objective cone would have to be multiplied by the number of affected Restore terms.

7 7. Restore-sensitive MultiTag normal form

The objective charges each active frame coordinate by at least `mu`, includes arbitrary nonnegative Tag-support terms, and charges the local Restore term by $t_R \geq 0$.

Deleting `k` frame coordinates in a sound zero-sum subsequence refunds at least `k mu`. By the incidence contract and Lemma 2 it adds at most `k(b-1)t_R` in Restore cost. Hence deletion is non-increasing whenever

$mu \geq (b-1)t_R$.

Theorem 3 (MultiTag support normal form). Under the registered grammar and incidence contract, if $mu \geq (b-1)t_R$, every admitted instance has an exact optimum satisfying

$$\mathrm{support}(R) \leq \mathrm{zsf}(H_R; A_R) \leq \mathrm{rank}(H_R) \leq s+1$$

simultaneously for every frame `R`.

Proof. Zero signature preserves the partner and Tag constraints by the registered semantic map. The objective calculation above establishes deletion dominance. Theorem 1 then gives the simultaneous zsf ceiling. Because `H_R` is an elementary binary subgroup of `F_2^(s+1)`, Section 4 gives the rank bounds. □

The first inequality may be strictly stronger than rank when the fixed admissible alphabet does not realize a basis of its generated subgroup.

8 8. Sharp R6M control

For the frozen one-Tag, three-block R6M family,

$b = 3$, $s = 1$, $mu = 2$, $t_R = 1$.

The objective lies on the cone boundary. The registered signature alphabet realizes a basis of `F_2^2`, so the rank-only deletion ceiling is two. The commit-bound parent evidence supplies the all-size upper result and the independent support-one obstruction needed to conclude

$kappa_R 6M = 2$,

where *kappa* denotes the mathematical minimum uniform support bound of the compiler family under the frozen objective. This exact value is a compiler result; the zero-sum/rank argument alone would not establish it.

No corresponding lower theorem is claimed for general MultiTag-TARE.

9 9. Relation to prior work

Zero-sum sequence theory owns Davenport-type thresholds and restricted-alphabet variants. Sparse integer optimization owns general support bounds for optimal solutions. Standard stabilizer formalism owns the binary symplectic signature machinery. Proof-complexity and formal-methods literatures own the general distinction between a proof system and the object it certifies.

The residual contribution is narrower: the registered TARE/MultiTag semantic signature, the exact Restore sensitivity and objective cone, the whole-instance simultaneous deletion contract, and the sharp R6M control.

10 10. Reproducibility and authority

The R6M parent theorem, objective ledger and necessity witness are commit-bound in the existing A1 evidence package. The finite verifiers test the local Restore function and small-group signature mechanisms, but all-size authority comes from the proofs above and from the cited R6M parent theorem. Internal independent implementations are not external replication.

11 11. Limitations

1. The zero-sum object is donor mathematics; novelty is not claimed for zsf itself.
2. The deletion theorem requires whole-instance soundness, objective dominance and support monotonicity.
3. The MultiTag cone uses the explicitly stated one-deletion/one-Restore-argument incidence contract.
4. Outside $mu \geq (b-1)t_R$, the proof is silent; no larger-support necessity follows.
5. General MultiTag sharpness is open.
6. Structural support is not T count, depth, runtime, qubits, or quantum advantage.
7. Submission-date donor overlap and independent specialist proof review remain external scientific checks.

12 12. Conclusion

The scientifically defensible result is not a new zero-sum invariant. It is a compiler theorem specifying exactly when donor zero-sum mathematics can be converted into a simultaneous support normal form. The global fixed-point proof closes the multi-generator quantifier; the instance-level alphabet removes solution-relative circularity; the Restore incidence and exact `b-1` sensitivity expose the objective condition; and R6M shows one case where the certificate ceiling is genuinely intrinsic. That bounded result is the journal object.

13 Tool-use disclosure

A generative language model assisted manuscript organization, language revision, adversarial review, and submission-package preparation. The listed author remains responsible for the mathematical statements, proofs, references, executable claims, and final text.

14 Data and code availability

The source package accompanying this manuscript contains finite-group and Restore-sensitivity control records used for implementation checks. These files are provenance aids rather than theorem authority; the displayed proofs and the cited parent theorem/witness carry the all-size claims.

15 Author contributions

Sze Chun Yiu is the sole listed author and performed the authorship contributions represented in this manuscript, including the formal analysis, computational verification, and manuscript preparation.

16 References

1. N. Schillo, A. Sturm, and R. Quay, "TARE: Block Encoding Linear Combinations of Pauli Strings Without Ancilla State Preparation," arXiv:2601.05740v4 [quant-ph] (2026).

2. I. Aliev, J. A. De Loera, F. Eisenbrand, T. Oertel, and R. Weismantel, "The Support of Integer Optimal Solutions," *SIAM Journal on Optimization* **28**, 2152–2157 (2018). DOI: 10.1137/17M1162792.

3. M. Freeze and W. A. Schmid, "Remarks on a generalization of the Davenport constant," *Discrete Mathematics* **310**, 3373–3389 (2010). DOI: 10.1016/j.disc.2010.07.028.

4. G. Wang, "The universal zero-sum invariant and weighted zero-sum for infinite abelian groups," *Communications in Algebra* **53**(4), 1581–1599 (2025). DOI: 10.1080/00927872.2024.2418017.

5. J. Dehaene and B. De Moor, "Clifford group, stabilizer states, and linear and quadratic operations over GF(2)," *Physical Review A* **68**, 042318 (2003). DOI: 10.1103/PhysRevA.68.042318.

6. S. Aaronson and D. Gottesman, "Improved simulation of stabilizer circuits," *Physical Review A* **70**, 052328 (2004). DOI: 10.1103/PhysRevA.70.052328.